In [1]:
```python
import pandas as pd
```

In [2]:
```python
import seaborn as sns
```

In [3]:
```python
import numpy as np
```

In [4]:
```python
import matplotlib.pyplot as plt
```

In [5]:
```python
df=pd.read_csv("academic_performance.csv")
```

In [6]:
```python
#incosistencies
for col in ['Maths','Science','English','Attendance']:
    print("Incosistent data in",col)
    print(df[(df[col]<0) | (df[col]>100)])
```

```
Incosistent data in Maths
    Student_ID  Maths  Science  English  Attendance
6            7  150.0     85.0     65.0        88.0
20          21  110.0     85.0     88.0        90.0
Incosistent data in Science
   Student_ID  Maths  Science  English  Attendance
7           8   72.0    400.0     78.0        92.0
Incosistent data in English
Empty DataFrame
Columns: [Student_ID, Maths, Science, English, Attendance]
Index: []
Incosistent data in Attendance
    Student_ID  Maths  Science  English  Attendance
17          18   92.0     95.0     98.0       105.0
```

In [7]:
```python
for col in ['Maths','Science','English','Attendance']:
    df.loc[(df[col]<0) | (df[col]>100),col]=np.nan
```

In [8]:
```python
df
```

Out[8]:

	Student_ID	Maths	Science	English	Attendance
0	1	78.0	82.0	75.0	85.0
1	2	85.0	79.0	80.0	90.0
2	3	NaN	88.0	70.0	78.0
3	4	95.0	NaN	85.0	95.0
4	5	40.0	35.0	NaN	60.0
5	6	88.0	90.0	88.0	NaN
6	7	NaN	85.0	65.0	88.0
7	8	72.0	NaN	78.0	92.0
8	9	66.0	70.0	68.0	75.0
9	10	55.0	60.0	58.0	65.0
10	11	90.0	92.0	94.0	98.0
11	12	48.0	52.0	50.0	55.0
12	13	82.0	78.0	80.0	82.0
13	14	76.0	NaN	74.0	80.0
14	15	NaN	65.0	60.0	70.0
15	16	68.0	72.0	70.0	77.0
16	17	85.0	88.0	90.0	89.0
17	18	92.0	95.0	98.0	NaN
18	19	35.0	30.0	40.0	45.0
19	20	60.0	58.0	62.0	68.0
20	21	NaN	85.0	88.0	90.0
21	22	73.0	75.0	72.0	NaN

	Student_ID	Maths	Science	English	Attendance
22	23	80.0	82.0	78.0	85.0
23	24	65.0	67.0	70.0	72.0
24	25	90.0	88.0	NaN	95.0
25	26	55.0	50.0	48.0	60.0
26	27	78.0	76.0	80.0	83.0
27	28	82.0	85.0	88.0	87.0
28	29	95.0	98.0	96.0	99.0
29	30	20.0	25.0	30.0	35.0

In [9]:
```
#column-wise inconsitent values
df.isnull().sum()
```

Out[9]:
```
Student_ID    0
Maths         4
Science       3
English       2
Attendance    3
dtype: int64
```

In [10]:
```
#replace null values with 0
df.fillna(0)
```

Out[10]:

	Student_ID	Maths	Science	English	Attendance
0	1	78.0	82.0	75.0	85.0
1	2	85.0	79.0	80.0	90.0
2	3	0.0	88.0	70.0	78.0
3	4	95.0	0.0	85.0	95.0
4	5	40.0	35.0	0.0	60.0
5	6	88.0	90.0	88.0	0.0
6	7	0.0	85.0	65.0	88.0
7	8	72.0	0.0	78.0	92.0
8	9	66.0	70.0	68.0	75.0
9	10	55.0	60.0	58.0	65.0
10	11	90.0	92.0	94.0	98.0
11	12	48.0	52.0	50.0	55.0
12	13	82.0	78.0	80.0	82.0
13	14	76.0	0.0	74.0	80.0
14	15	0.0	65.0	60.0	70.0
15	16	68.0	72.0	70.0	77.0
16	17	85.0	88.0	90.0	89.0
17	18	92.0	95.0	98.0	0.0
18	19	35.0	30.0	40.0	45.0
19	20	60.0	58.0	62.0	68.0
20	21	0.0	85.0	88.0	90.0
21	22	73.0	75.0	72.0	0.0

	Student_ID	Maths	Science	English	Attendance
22	23	80.0	82.0	78.0	85.0
23	24	65.0	67.0	70.0	72.0
24	25	90.0	88.0	0.0	95.0
25	26	55.0	50.0	48.0	60.0
26	27	78.0	76.0	80.0	83.0
27	28	82.0	85.0	88.0	87.0
28	29	95.0	98.0	96.0	99.0
29	30	20.0	25.0	30.0	35.0

In [11]:
```
#repplacing Nan with NULL
df.replace(np.nan,"NUll")
```

Out[11]:

	Student_ID	Maths	Science	English	Attendance
0	1	78.0	82.0	75.0	85.0
1	2	85.0	79.0	80.0	90.0
2	3	NUll	88.0	70.0	78.0
3	4	95.0	NUll	85.0	95.0
4	5	40.0	35.0	NUll	60.0
5	6	88.0	90.0	88.0	NUll
6	7	NUll	85.0	65.0	88.0
7	8	72.0	NUll	78.0	92.0
8	9	66.0	70.0	68.0	75.0
9	10	55.0	60.0	58.0	65.0
10	11	90.0	92.0	94.0	98.0
11	12	48.0	52.0	50.0	55.0
12	13	82.0	78.0	80.0	82.0
13	14	76.0	NUll	74.0	80.0
14	15	NUll	65.0	60.0	70.0
15	16	68.0	72.0	70.0	77.0
16	17	85.0	88.0	90.0	89.0
17	18	92.0	95.0	98.0	NUll
18	19	35.0	30.0	40.0	45.0
19	20	60.0	58.0	62.0	68.0
20	21	NUll	85.0	88.0	90.0
21	22	73.0	75.0	72.0	NUll

	Student_ID	Maths	Science	English	Attendance
22	23	80.0	82.0	78.0	85.0
23	24	65.0	67.0	70.0	72.0
24	25	90.0	88.0	NUll	95.0
25	26	55.0	50.0	48.0	60.0
26	27	78.0	76.0	80.0	83.0
27	28	82.0	85.0	88.0	87.0
28	29	95.0	98.0	96.0	99.0
29	30	20.0	25.0	30.0	35.0

In [12]:
```
df["Maths"].mean()
```

Out[12]:
```
71.26923076923077
```

In [13]:
```
#replacing NaN with mean of column
df["Maths"]=df["Maths"].fillna(df["Maths"].mean()).round(2)
```

In [14]:
```
df
```

Out[14]:

	Student_ID	Maths	Science	English	Attendance
0	1	78.00	82.0	75.0	85.0
1	2	85.00	79.0	80.0	90.0
2	3	71.27	88.0	70.0	78.0
3	4	95.00	NaN	85.0	95.0
4	5	40.00	35.0	NaN	60.0
5	6	88.00	90.0	88.0	NaN
6	7	71.27	85.0	65.0	88.0
7	8	72.00	NaN	78.0	92.0
8	9	66.00	70.0	68.0	75.0
9	10	55.00	60.0	58.0	65.0
10	11	90.00	92.0	94.0	98.0
11	12	48.00	52.0	50.0	55.0
12	13	82.00	78.0	80.0	82.0
13	14	76.00	NaN	74.0	80.0
14	15	71.27	65.0	60.0	70.0
15	16	68.00	72.0	70.0	77.0
16	17	85.00	88.0	90.0	89.0
17	18	92.00	95.0	98.0	NaN
18	19	35.00	30.0	40.0	45.0
19	20	60.00	58.0	62.0	68.0
20	21	71.27	85.0	88.0	90.0
21	22	73.00	75.0	72.0	NaN

	Student_ID	Maths	Science	English	Attendance
22	23	80.00	82.0	78.0	85.0
23	24	65.00	67.0	70.0	72.0
24	25	90.00	88.0	NaN	95.0
25	26	55.00	50.0	48.0	60.0
26	27	78.00	76.0	80.0	83.0
27	28	82.00	85.0	88.0	87.0
28	29	95.00	98.0	96.0	99.0
29	30	20.00	25.0	30.0	35.0

In [15]:
```
#replacing NaN with mean of column
for col in ['Maths','Science','English','Attendance']:
    df[col]=df[col].fillna(df[col].mean()).round(2)
```

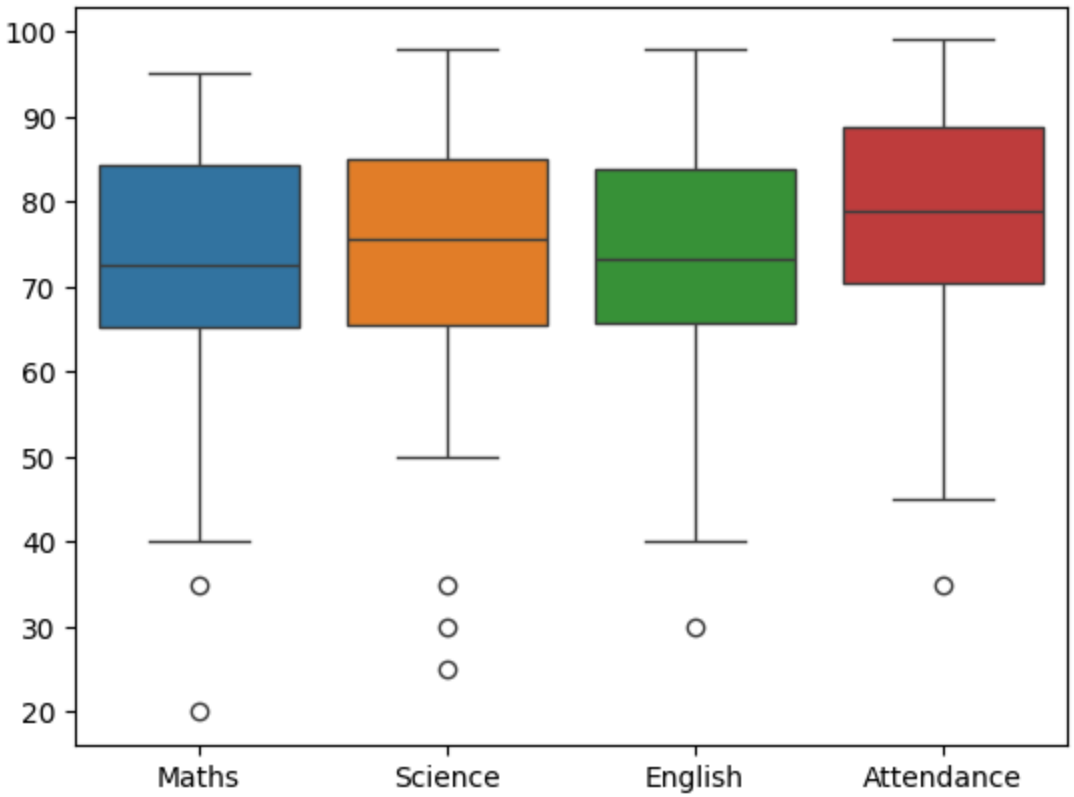
In [16]:
```
df
```

Out[16]:

	Student_ID	Maths	Science	English	Attendance
0	1	78.00	82.00	75.00	85.0
1	2	85.00	79.00	80.00	90.0
2	3	71.27	88.00	70.00	78.0
3	4	95.00	72.22	85.00	95.0
4	5	40.00	35.00	72.68	60.0
5	6	88.00	90.00	88.00	77.7
6	7	71.27	85.00	65.00	88.0
7	8	72.00	72.22	78.00	92.0
8	9	66.00	70.00	68.00	75.0
9	10	55.00	60.00	58.00	65.0
10	11	90.00	92.00	94.00	98.0
11	12	48.00	52.00	50.00	55.0
12	13	82.00	78.00	80.00	82.0
13	14	76.00	72.22	74.00	80.0
14	15	71.27	65.00	60.00	70.0
15	16	68.00	72.00	70.00	77.0
16	17	85.00	88.00	90.00	89.0
17	18	92.00	95.00	98.00	77.7
18	19	35.00	30.00	40.00	45.0
19	20	60.00	58.00	62.00	68.0
20	21	71.27	85.00	88.00	90.0
21	22	73.00	75.00	72.00	77.7

	Student_ID	Maths	Science	English	Attendance
22	23	80.00	82.00	78.00	85.0
23	24	65.00	67.00	70.00	72.0
24	25	90.00	88.00	72.68	95.0
25	26	55.00	50.00	48.00	60.0
26	27	78.00	76.00	80.00	83.0
27	28	82.00	85.00	88.00	87.0
28	29	95.00	98.00	96.00	99.0
29	30	20.00	25.00	30.00	35.0

In [17]:
```
sns.boxplot(data=df[['Maths','Science','English','Attendance']])
plt.show()
```



In [18]:
```
Q1=df['Maths'].quantile(0.25)
Q2=df['Maths'].quantile(0.50)
Q3=df['Maths'].quantile(0.75)
```

In [19]:
```
Q1
```

Out[19]: 65.25

In [20]:
```
Q2
```

Out[20]: 72.5

In [21]:
```
Q3
```

Out[21]: 84.25

In [22]:
```
IQR=Q3-Q1
lower=Q1-1.5*IQR
print("IQR",IQR)
print("LOW",lower)
upper=Q3+1.5*IQR
print("Upper",upper)
```

```
IQR 19.0
LOW 36.75
Upper 112.75
```

In [23]:
```
cols=['Maths','Science','English','Attendance']
```

In [24]:
```
all_Q1=df[cols].quantile(0.25)
```

In [25]:
```
all_Q1
```

Out[25]:
```
Maths         65.25
Science       65.50
English       65.75
Attendance    70.50
Name: 0.25, dtype: float64
```

In [26]:
```
all_Q3=df[cols].quantile(0.75)
```

In [27]:
```
all_Q3
```

Out[27]:
```
Maths         84.25
Science       85.00
English       83.75
Attendance    88.75
Name: 0.75, dtype: float64
```

In [28]:
```
all_IQR=all_Q3-all_Q1
```

In [29]:
```
all_IQR
```

Out[29]:
```
Maths         19.00
Science       19.50
English       18.00
Attendance    18.25
dtype: float64
```

In [30]:
```
lower_bound=all_Q1-1.5*all_IQR
```

In [31]:
```
lower_bound
```

Out[31]:
```
Maths         36.750
Science       36.250
English       38.750
Attendance    43.125
dtype: float64
```

In [32]:
```
upper_bound=all_Q3+1.5*all_IQR
```

In [33]:
```
upper_bound
```

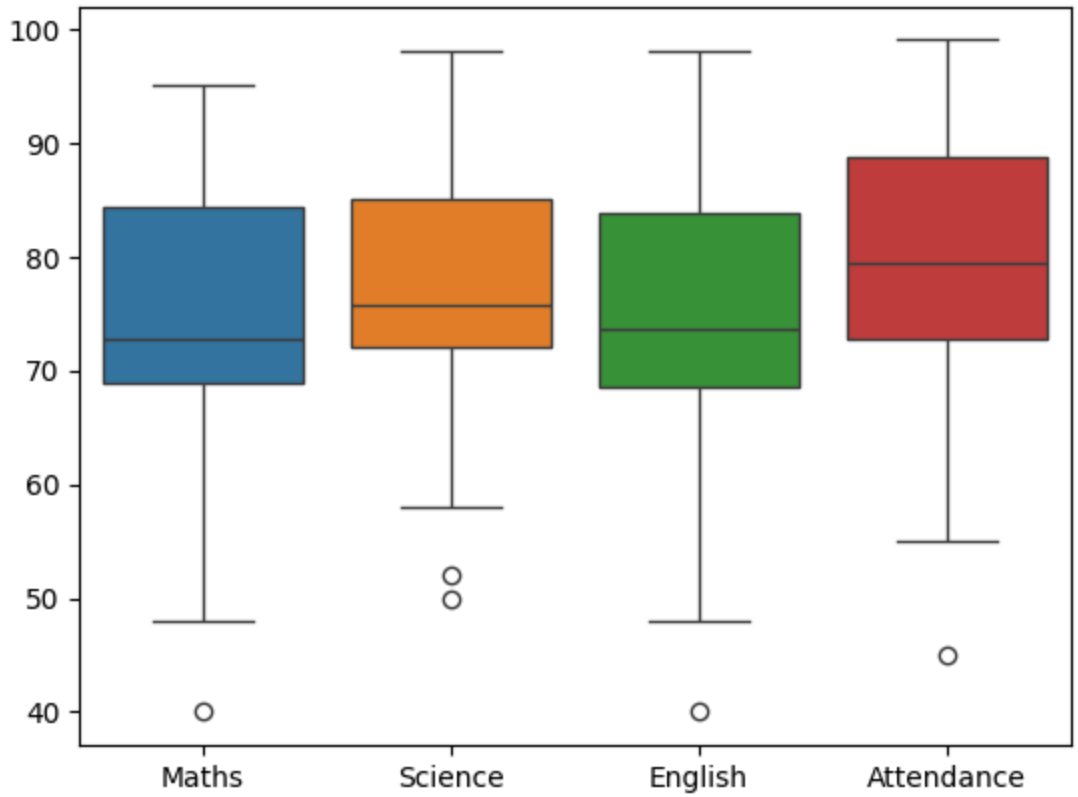
Out[33]:
```
Maths         112.750
Science       114.250
English       110.750
Attendance    116.125
dtype: float64
```

In [34]:
```
#replace with median
for col in ['Maths','Science','English','Attendance']:
   df.loc[(df[col]<lower_bound[col]) | (df[col]>upper_bound[col]),col]=df[col].median()
```

In [35]:
```
#sns.boxplot(data=df[['Maths','Science','English','Attendance']])
plt.show()
```

In [36]:
```
for col in ['Maths','Science','English','Attendance']:
    df.loc[(df[col]<lower_bound[col]),col]=lower_bound[col]
    df.loc[(df[col]>upper_bound[col]),col]=upper_bound[col]
```

In [37]:
```
sns.boxplot(data=df[['Maths','Science','English','Attendance']])
plt.show()
```



Transformation

In [38]:
```
#skewness
mt.hist(df['Maths'],bins=5)
mt.title("Maths before Transformation")
mt.show()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[38], line 2
      1 #skewness
----> 2 mt.hist(df['Maths'],bins=5)
      3 mt.title("Maths before Transformation")
      4 mt.show()

NameError: name 'mt' is not defined
```

In [ ]:
```
#negative skewness
df['Maths'].skew()
```

In [ ]:
```
df['Maths Transformed']=np.log(df['Maths'])
```

In [ ]:
```
df['Maths Transformed'].skew()
```

In [ ]:
```
max=df['Maths'].max()
df['Maths reflected']=max+1-df['Maths']
df['Maths Transformed']=np.sqrt(df['Maths reflected'])
```

In [ ]:
```
df['Maths Transformed'].skew()
```

In [ ]:
```
mt.hist(df['Maths Transformed'],bins=5)
mt.title("Maths after Transformation")
mt.show()
```

In [ ]: